Assignment: Static Routing vs Dynamic Routing (RIP)

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Task 1

Question:

List three differences between static routing and dynamic routing in a table format, using real-world analogies from apps like Google Maps or Swiggy to make each point relatable.

Answer:

STATIC ROUTING	DYNAMIC ROUTING	REAL-WORLD ANALOGY
ROUTES ARE CONFIGURED MANUALLY BY THE NETWORK ADMINISTRATOR.	Routes are learned and updated automatically using routing protocols such as RIP.	Google Maps: Static routing is like always following the same road to reach your destination. Dynamic routing is like Google Maps automatically suggesting a better route when there is traffic.
STATIC ROUTES DO NOT CHANGE AUTOMATICALLY WHEN THE NETWORK CHANGES.	Dynamic routes automatically update whenever there is a change in the network.	Swiggy: If a delivery partner always follows the same route, it is similar to static routing. If the app finds another route because a road is closed, it is similar to dynamic routing.
STATIC ROUTING IS BEST SUITED FOR SMALL AND STABLE NETWORKS.	Dynamic routing is best suited for large and frequently changing networks.	A small shop with only one delivery route can use static routing, while a company with many branches uses dynamic routing to automatically find the best route.

Task 2

Question:

Open Cisco Packet Tracer and load a pre-configured network file with three routers using RIP. Use the 'show ip route' and 'show ip protocols' commands on each router to observe and write down the RIP-learned routes and their metrics.

Answer:

Commands Used

show ip route

show ip protocols

Observation Table

ROUTER	RIP LEARNED ROUTE	METRIC (HOP COUNT)
ROUTER1	192.168.2.0/24	1
ROUTER1	192.168.3.0/24	2
ROUTER2	192.168.1.0/24	1
ROUTER2	192.168.3.0/24	1
ROUTER3	192.168.1.0/24	2
ROUTER3	192.168.2.0/24	1

Observation

After executing the show ip route command, the routing table displayed routes learned through RIP, identified by the letter R. The show ip protocols command displayed the RIP configuration, update timers, and participating networks. RIP selected the path with the lowest hop count as the best route.

Task 3

Question:

Simulate a network change by shutting down one interface on a router in the RIP-enabled topology. Observe and record how long it takes for the routing tables on the other routers to update and remove the unreachable route.

Hint: Use the 'shutdown' command on an interface and monitor with 'show ip route' repeatedly.

Answer:

Commands Used

Router(config)# interface g0/0

Router(config-if)# shutdown

Verification Command

show ip route

Observation

After shutting down the router interface, the connected network became unavailable. RIP automatically informed the neighboring routers about the network change. The unreachable route was removed from the routing tables after approximately 180 seconds (3 minutes) because RIP's default invalid timer expired. This demonstrated that RIP automatically updates routing information whenever a network path changes.

Task 4

Question:

Explain in 3–4 sentences how dynamic routing in apps like Uber or Zomato delivery tracking is similar to RIP dynamic routing in networks. Focus on how routes or paths are updated automatically in both cases.

Answer:

Dynamic routing in computer networks works similarly to navigation used by Uber or Zomato. If a road becomes blocked or traffic increases, the application automatically finds a better route for the driver. In the same way, RIP automatically updates routing tables whenever a network path changes or becomes unavailable. Both systems help ensure that the destination is reached through the best available path without requiring manual changes.

Conclusion

Static routing is simple and suitable for small networks, but it requires manual configuration whenever changes occur. Dynamic routing protocols such as RIP automatically learn and update routes, making them more suitable for large and changing networks. Practical activities performed in Cisco Packet Tracer demonstrated how RIP updates routing tables automatically after a network change, ensuring continuous communication between devices.